
MODEL SELECTION AND VALIDATION REPORT (remove_6_biomass model)

OBSERVATIONS:1692

SANITY: OK - all models in the selection process passed sanity checks.

SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

MODEL SUMMARY (remove_6_biomass model)

Spatiotemporal model fit by REML ['sdmTMB']

Formula: fixed_formula

Time column: year

Family: lognormal(link = 'log')

Conditional model:

	coef.est	coef.se
(Intercept)	5.62	0.19
remove_6_kde_fric	-0.28	0.04
remove_6_kde_feve	-0.23	0.03
sbt_mean	0.25	0.11
sbt_sd	0.02	0.05
sst_sd	0.01	0.05
log_chla_mean	0.04	0.04
depth	0.03	0.05
log_sum_fishing	0.07	0.04

Dispersion parameter: 1.06

Spatiotemporal AR1 correlation (rho): 0.93

Matérn range: 190.03

Spatiotemporal marginal AR1 SD: 1.00

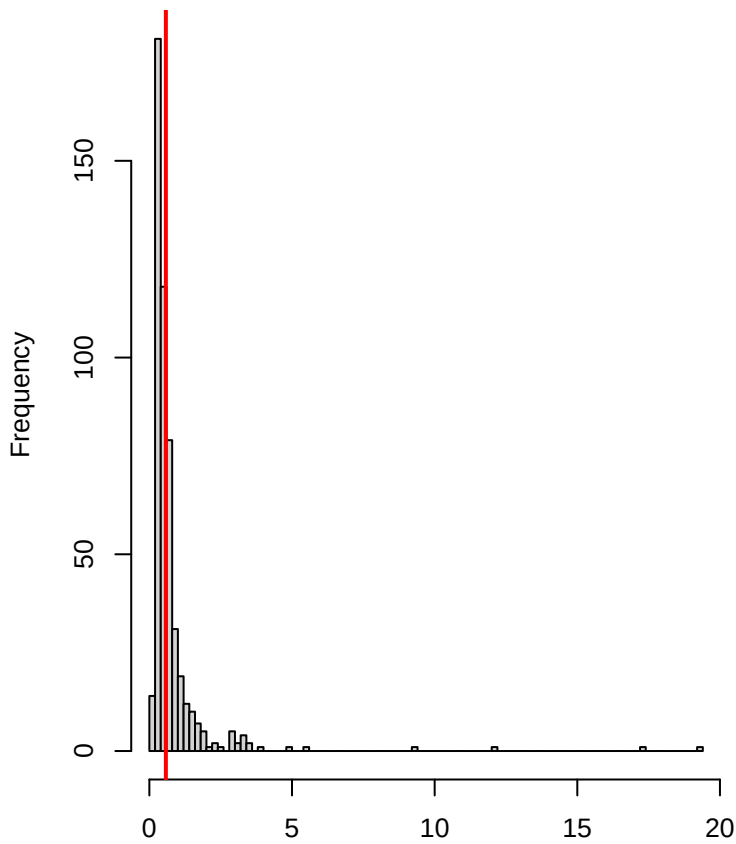
REML criterion at convergence: 11062.148

See ?tidy.sdmTMB to extract these values as a data frame.

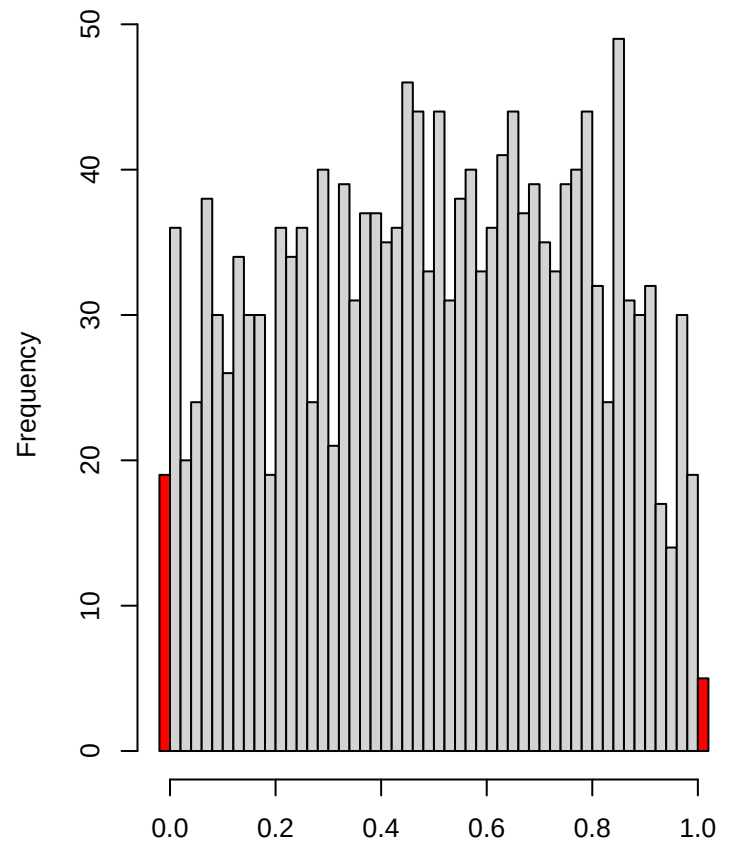
SELECTION STATISTICS

	df	AIC	delta_AIC
spatiotemporal_random_fields_ar1	13	22150.30	0.2902807
spatiotemporal_random_fields_ar1_with_anisotropy	15	22150.59	68.2568891
spatial_random_fields	12	22218.84	3.4898163
spatial_random_fields_with_anisotropy	14	22222.33	133.3411035
spatiotemporal_random_fields_iid_with_anisotropy	14	22355.67	7.2761879
spatiotemporal_random_fields_iid	12	22362.95	124.3034099
error_only	10	22487.25	NA

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated



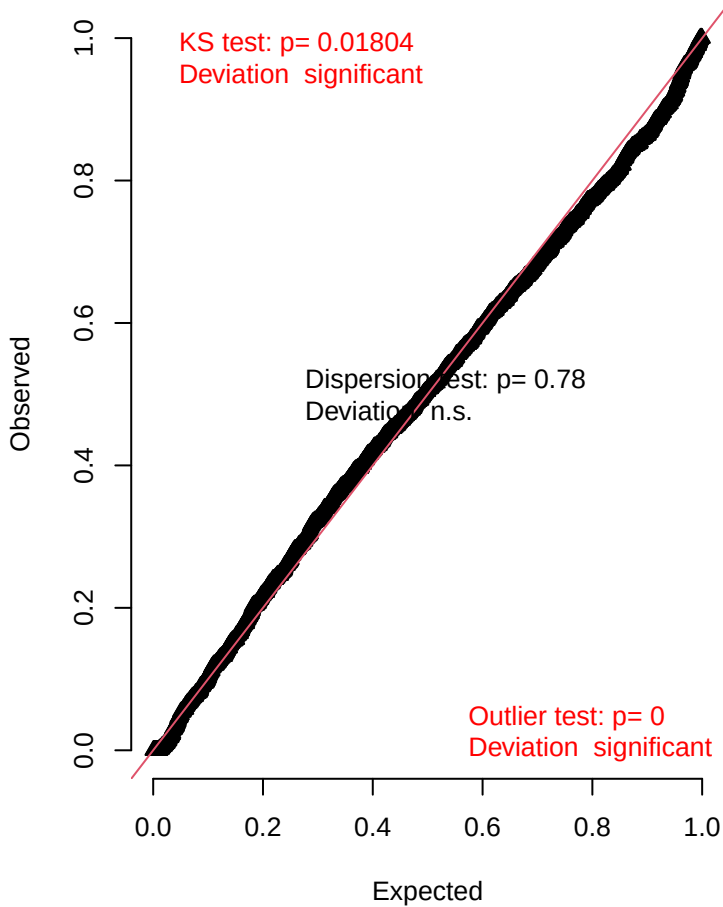
Outlier test significant



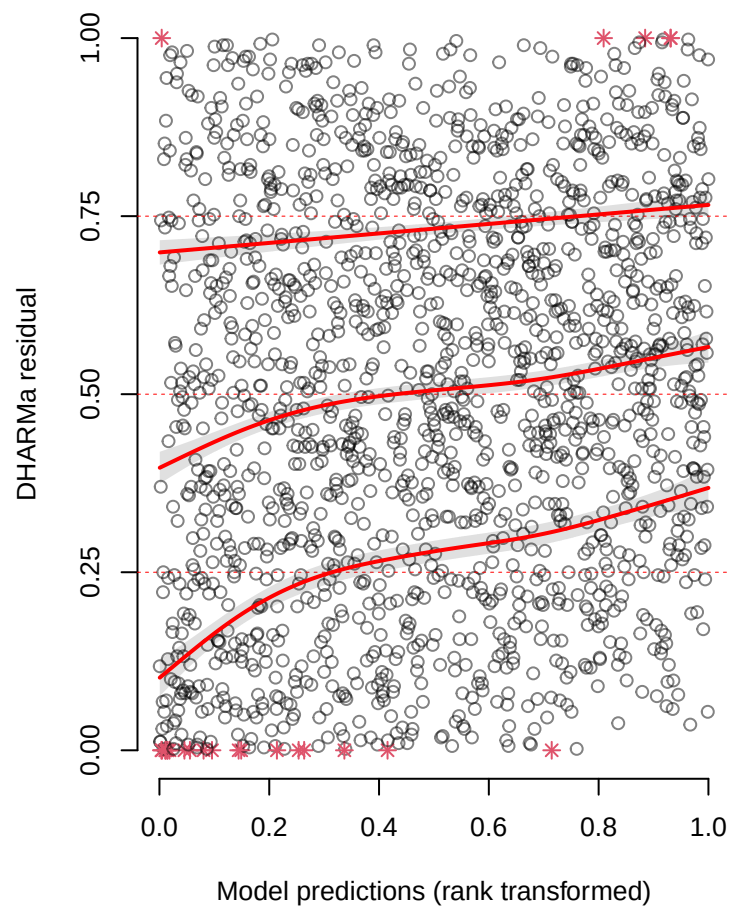
Simulated values, red line = fitted model. p-value (two.sided) :

Residuals (outliers are marked red)

QQ plot residuals



DHARMa residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



DHARMa Moran's I test for distance-based autocorrelation
Moran's I = 0.024 | H1: Distance-based autocorrelation | p-value = 0.009

